Datawarehouse

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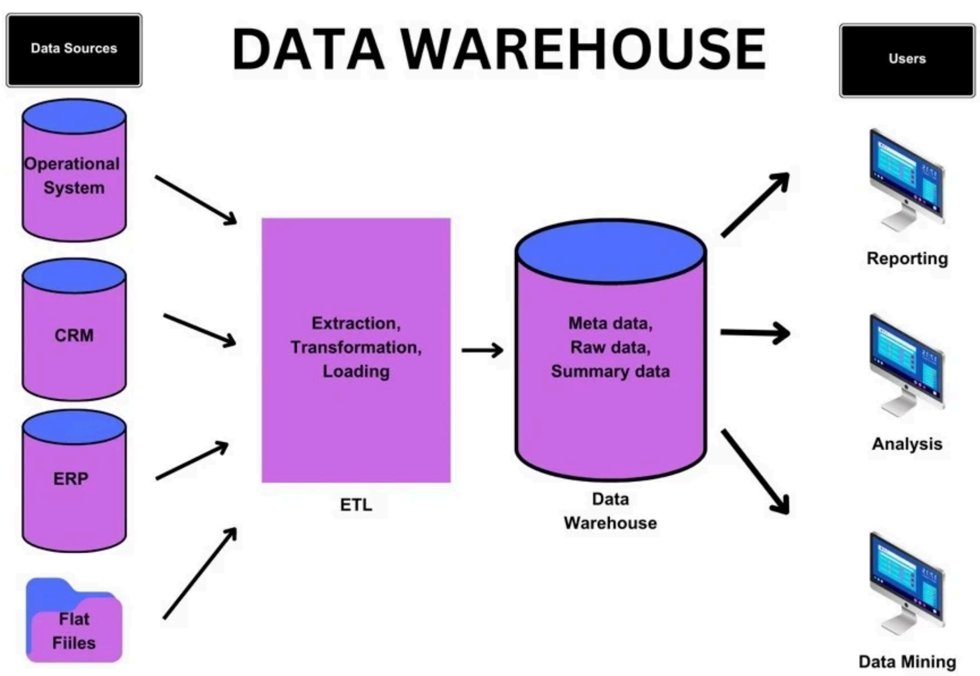
Datawarehouse:

Datawarehouse is centralized data system that stores the structured historical data collected from various disparate data sources across an organization.

Datawarehouse enables to study the data and provide the insights to improve the business.

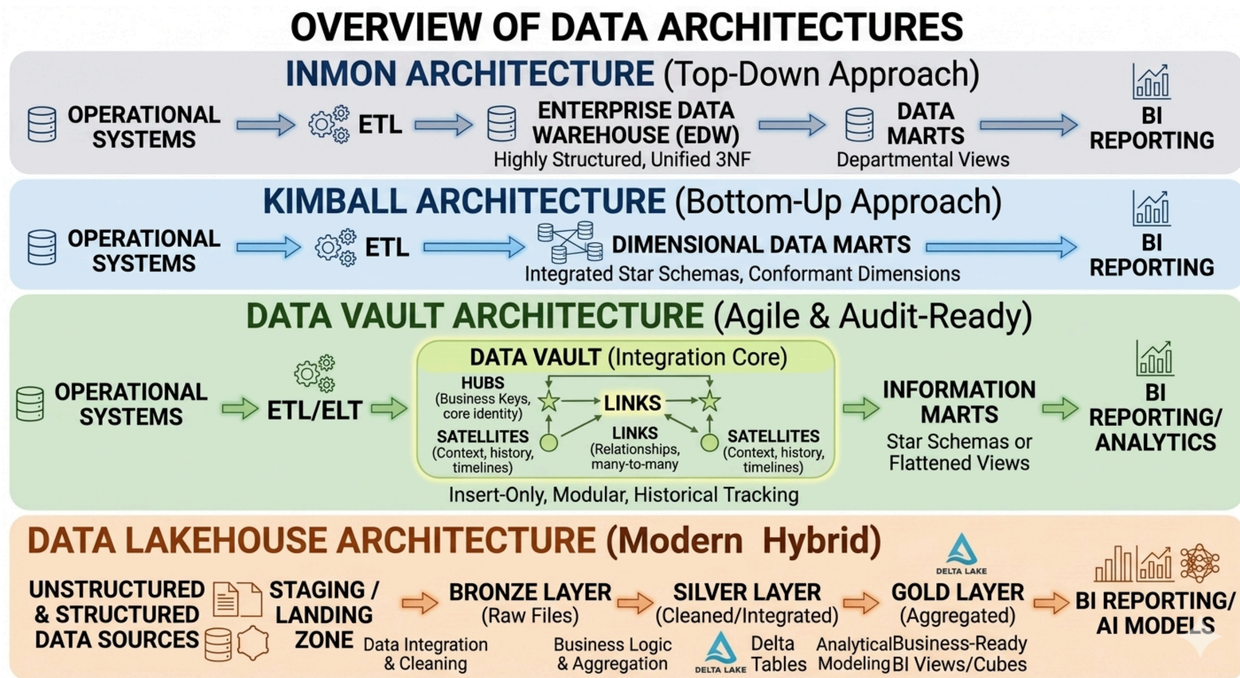
Reporting purpose.

Snowflake, Big query,Amazon redshift are the best examples of datawarehouse systems.



There are different architectures to design the datawarehouse.

1. Inmon Architecture
2. Kimball Architecture
3. Data vault
4. Medallion Architecture



1. Inmon Architecture

This approach builds a single, highly organized central database first, then breaks it down for users.

- **Operational Systems:** The original apps and databases where data is created.
- **ETL:** The software process that pulls data out, cleans it, and moves it.
- **Enterprise Data Warehouse (EDW):** One single, massive central database. It organizes data using a strict rule called **3NF** to ensure there are zero duplicates.
- **Data Marts:** Smaller databases that copy specific parts of the data from the central warehouse. Each department gets its own view.
- **BI Reporting:** The final screens, charts, and reports used by business staff to read the data.

2. Kimball Architecture

This approach focuses on building smaller, department-specific databases first, which then connect to form the whole system.

- **Operational Systems & ETL:** The data is created and moved just like in the Inmon system.
- **Dimensional Data Marts:** Instead of one massive central database, data goes straight into smaller, specialized databases. They use a structure called **Star Schemas** to make searching fast. They also use **Conformant Dimensions**, meaning every department agrees on the exact same definitions for key terms.
- **BI Reporting:** Business users directly read these data marts to generate their daily reports.

3. Data Vault Architecture

This approach is built to handle frequent business changes and keep a perfect record of every change over time.

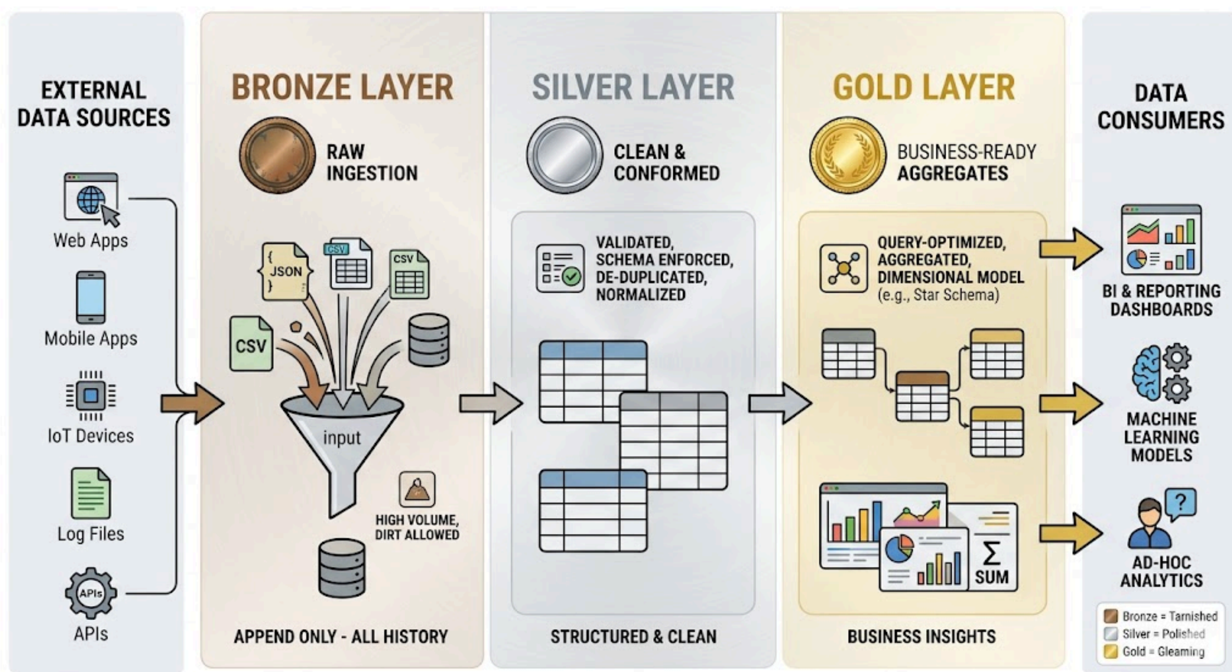
- **ETL/ELT:** The process that moves the data into the system.
- **Data Vault (Integration Core):** The central storage area that separates data into three specific types of tables:
    - **Hubs:** Tables that hold unique business identification keys.
    - **Links:** Tables that connect those keys to show relationships.
    - **Satellites:** Tables that hold descriptions and details. This part is **Insert-Only**, meaning old data is never deleted or overwritten; new rows are simply added to keep a full history.
- **Information Marts:** Simplified tables built on top of the vault to turn the complex history into standard, readable views.
- **BI Reporting / Analytics:** The final tools used to read and analyze these simplified views.

4. Data Lakehouse Architecture(Medallion Architecture)

This is a modern system designed to handle both standard text/numbers and messy, unstructured data (like files and images) in a multi-step pipeline.

- **Unstructured & Structured Data Sources:** All types of data entering the system.
- **Staging / Landing Zone:** The initial drop-off point where data arrives exactly as it is.
- **Bronze Layer:** Storage for the raw, unedited files.
- **Silver Layer:** Storage where the data is cleaned, filtered, and organized into structured **Delta Tables**.
- **Gold Layer:** Storage where the cleaned data is grouped, calculated, and optimized for final business use.
- **BI Reporting / AI Models:** The final output layer used for creating business charts and training artificial intelligence models.

Modern data warehouses are building using medallion architecture.
In a modern warehouse setup, instead of separating data by file folders in a cloud storage bucket, the architecture is typically implemented using separate Databases or Schemas within the same warehouse.



1. Bronze (The "Landing" or "Staging" Schema):

- Automated ingestion tools (like Fivetran, Airbyte, or Kafka) dump data directly into the warehouse. No cleaning, zero transformations.
- Data is often loaded as-is into raw tables. For data like JSON or API payloads, warehouses use native semi-structured columns

2. Silver (The "Base" or "Integration" Schema):

- This is where dbt (data build tool) or SQL stored procedures take over
- The data is deduplicated, null values are handled, and tables are joined to create clean "Enterprise Views”

3. Gold Layer (The Business-Ready Aggregates):

- data here is structured specifically to power analytics, machine learning models, or executive dashboards.
- We apply heavy business logic, aggregations, and optimizations (like star schemas or dimensional modeling).

ETL(Extract, Transform, Load):

ETL extracts raw data from various systems, transforms it for business logic alignment, and loads it into a Data Warehouse to enable strategic decision-making.

1. Extract(E):

collects data from various source systems, including databases, spreadsheets, cloud-based platforms, and files. The data is in its raw form.

Extraction Methods:

- **Pull Extraction:** The ETL pipeline actively requests/queries data from the source system at scheduled intervals.
- **Push Extraction:** The source system automatically sends or broadcasts data to the ETL target system as soon as it's available

Extraction types:

- **Full Extraction:** Pull the entire dataset from the source every single time, regardless of whether it has changed.
- **Incremental Extraction:** Only extract data that has changed or been created since the last extraction run (much more efficient).

2. Transform(T):

Transformation is where data is cleaned, reshaped, and prepared according to business rules.

Data Cleansing: The foundational **scrubbing** process. It includes:

- *Remove Duplicates:* Striking out redundant records.
- *Data Filtering:* Dropping irrelevant rows based on criteria.
- *Handling Missing Data / Invalid Values:* Deciding whether to drop nulls or impute them with default values.
- *Handling Unwanted Spaces:* Trimming leading/trailing whitespace.
- *Data Type Casting:* Ensuring numbers are floats/integers and dates are true datetime objects.
- *Outlier Detection:* Flagging or isolating extreme data anomalies.

Data Enrichment: Enhancing the dataset by appending external data or calculating new insights (e.g., adding geolocations based on a zip code).

Data Integration: Standardizing and combining data from different sources so they seamlessly talk to one another.

Derived Columns: Creating entirely new data fields calculated from existing ones

Data Normalization & Standardization: Structuring data to reduce redundancy and unifying formats (e.g., turning "Male", "M", and "male" into a uniform "M").

Business Rules & Logic: Applying unique company logic (e.g., classifying a customer as "VIP" if their spending exceeds a certain threshold).

Data Aggregations: Summarizing granular data into higher-level metrics (e.g., rolling up hourly transactions into a daily sales total).

3. Load(L):

Loading defines how the transformed data is written into the final repository

Processing Types:

- **Batch Processing:** Processing data in large blocks at scheduled intervals (e.g., every night at midnight).
- **Stream Processing:** Processing and loading data continuously, record by record, in near real-time.

Load Methods:

- **Full Load:** Overwriting the destination target entirely. Techniques include:
    - *Truncate & Insert:* Wiping the destination table clean and inserting all data fresh.
    - *Upsert:* Updating existing records if they match, and inserting them if they don't.
    - *Drop, Create, Insert:* Completely deleting the table structure from the database, recreating it, and filling it.
- **Incremental Load:** Appending new data to the existing target. Techniques include:
    - *Upsert:* Modifying changed rows and adding new ones.
    - *Append:* Blindly adding new rows to the bottom of the table without checking for existing ones.
    - *Merge:* Aligning target and source data to apply inserts, updates, or deletes dynamically.